

LAB NOC 04

Cisco Catalyst 2960 — STP Loop Detection & Prevention

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Ticket ID	NOC-LAB-04	NOC-LAB-04
Categoría / Category	STP — Detección y prevención de loops L2	STP — Layer 2 loop detection and prevention
Nivel / Level	CCNA — Switching / Spanning Tree Protocol	CCNA — Switching / Spanning Tree Protocol
Duración / Duration	Aprox. 45 minutos	Aprox. 45 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Diagnóstico de loops L2, análisis de syslog	L2 loop diagnosis, syslog analysis

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Crear un loop de Capa 2 en hardware real (Cisco Catalyst 2960), verificar que STP detecta y bloquea automáticamente el puerto redundante, documentar el incidente en formato NOC, remover el loop y confirmar la reconvergencia de STP mediante análisis de syslog.	Create a Layer 2 loop on real hardware (Cisco Catalyst 2960), verify that STP automatically detects and blocks the redundant port, document the incident in NOC format, remove the loop, and confirm STP reconvergence through syslog analysis.

INCIDENT TICKET — NOC-LAB-04

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-04
Priority / Prioridad	Medium
Service / Servicio	LAN Switching — Layer 2
Device / Equipo	Cisco Catalyst 2960 — Access Layer Switch
Symptoms / Sintomas	Inestabilidad de red, pérdida de paquetes intermitente, tiempos de respuesta lentos / Network instability, intermittent packet loss, slow response times
Initial Hypothesis	Posible loop de Capa 2 — STP podría estar bloqueando un puerto activamente / Possible Layer 2 loop — STP may be actively blocking a

Topology	port Fa0/6 conectado directamente a Fa0/16 en el mismo switch — path redundante / Fa0/6 directly connected to Fa0/16 on the same switch — redundant path
Root Cause	Loop de Capa 2 creado por cable directo Fa0/6 ↔ Fa0/16 / Layer 2 loop created by direct cable Fa0/6 ↔ Fa0/16
STP Action	STP coloco Fa0/16 en estado Blocking para prevenir broadcast storm / STP placed Fa0/16 in Blocking state to prevent broadcast storm
Resolution	Desconexion del cable de Fa0/16 — remocion del path redundante / Disconnection of Fa0/16 cable — removal of redundant path
Status	CERRADO — red estable, STP reconvergió / CLOSED — network stable, STP reconverged

PROCEDIMIENTO — 5 ETAPAS / PROCEDURE — 5 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Baseline: verificar estado inicial del switch	STAGE 1 — Baseline: verify initial switch state
2	show spanning-tree → identificar Root Bridge, roles de puertos y estados actuales	show spanning-tree → identify Root Bridge, port roles and current states
3	show etherchannel summary → verificar estado de bundles preexistentes	show etherchannel summary → verify status of pre-existing bundles
4	show vlan brief → confirmar VLANs activas y asignaciones	show vlan brief → confirm active VLANs and assignments
5	ETAPA 2 — Creación del loop: conectar cable Fa0/6 ↔ Fa0/16 en el mismo switch	STAGE 2 — Loop creation: connect cable Fa0/6 ↔ Fa0/16 on the same switch
6	Observar síntomas: inestabilidad de red, pérdida de paquetes, lentitud / Observe symptoms: network instability, packet loss, slowness	
7	ETAPA 3 — Diagnóstico: ejecutar show spanning-tree y confirmar hallazgos	STAGE 3 — Diagnosis: run show spanning-tree and confirm findings
8	Confirmar: Fa0/6 en estado Forwarding (Designated) — Fa0/16 en estado Blocking (Non-Designated)	Confirm: Fa0/6 in Forwarding state (Designated) — Fa0/16 in Blocking state (Non-Designated)
9	Confirmar: STP protegió la red — NO hubo broadcast storm gracias al protocolo	Confirm: STP protected the network — NO broadcast storm occurred thanks to the protocol
10	Ejecutar: show interfaces status, show mac address-table, show logging (metodología NOC completa)	Run: show interfaces status, show mac address-table, show logging (complete NOC methodology)
11	ETAPA 4 — Resolución: desconectar cable de Fa0/16 — remover path redundante	STAGE 4 — Resolution: disconnect cable from Fa0/16 — remove redundant path
12	ETAPA 5 — Verificación: show logging → confirmar LINEPROTO-5-UPDOWN en syslog	STAGE 5 — Verification: show logging → confirm LINEPROTO-5-UPDOWN in syslog

13	Verificar STP reconvergencia — Fa0/6 continua en Forwarding, red estable	Verify STP reconvergence — Fa0/6 continues in Forwarding, network stable
14	Cerrar ticket NOC-LAB-04 — incidente documentado y resuelto	Close ticket NOC-LAB-04 — incident documented and resolved

ESTADO STP OBSERVADO / OBSERVED STP STATE

PUERTO / PORT	ESTADO STP / STP STATE	ROL / ROLE	SIGNIFICADO / MEANING
Fa0/6	Forwarding	Designated	Trafico fluye normalmente / Traffic flows normally
Fa0/16	Blocking	Non-Designated	STP bloquea el puerto para evitar loop / STP blocked port to prevent loop

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show spanning-tree</code>	Muestra el estado STP: Root Bridge, roles y estados de puertos	Shows STP state: Root Bridge, port roles and states
<code>show interfaces status</code>	Muestra estado fisico de cada puerto (connected/notconnect/err-disabled)	Shows physical state of each port (connected/notconnect/err-disabled)
<code>show vlan brief</code>	Verifica VLANs activas y asignacion de puertos	Verifies active VLANs and port assignments
<code>show mac address-table</code>	Tabla MAC dinamica — util para rastrear origen del trafico	Dynamic MAC table — useful for tracking traffic source
<code>show logging</code>	Muestra syslog del switch — clave para reconstruir la secuencia del incidente	Shows switch syslog — key for reconstructing incident timeline
<code>show etherchannel summary</code>	Verifica estado de bundles EtherChannel (contexto de baseline)	Verifies EtherChannel bundle status (baseline context)

EVIDENCIA SYSLOG / SYSLOG EVIDENCE

#	MENSAJE / MESSAGE	LOG OUTPUT	INTERPRETACION / INTERPRETATION
1	LINEPROTO-5-UPDOWN	Interface FastEthernet0/16 changed state to down	Cable desconectado de Fa0/16 — path redundante removido / Fa0/16 cable disconnected — redundant path removed
2	LINEPROTO-5-UPDOWN	Interface FastEthernet0/6 changed state to down	Fa0/6 tambien bajo (sin destino — loop eliminado) / Fa0/6 also went down (no destination — loop eliminated)
3	STP converge	STP reconverged — Fa0/6 returns to Forwarding	Red estabilizada — unico path activo / Network stabilized — single active path

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
Loop L2 entre Fa0/6 y Fa0/16 causo inestabilidad de red (síntomas reportados por usuarios). / L2 loop between Fa0/6 and Fa0/16 caused network instability (symptoms reported by users).	STP detecto el loop automaticamente y coloco Fa0/16 en Blocking. Red no colapso. No requirio intervencion manual inmediata. / STP automatically detected the loop and placed Fa0/16 in Blocking. Network did not collapse. No immediate manual intervention required.
Durante el diagnostico se verifico que STP estaba activo y funcionando correctamente — es facil asumir que STP fallo cuando en realidad lo estaba manejando. / During diagnosis it was verified that STP was active and working correctly — it's easy to assume STP failed when it was actually handling it.	Regla NOC: antes de intervenir, leer el estado de STP completo. Si un puerto esta en Blocking es porque STP hizo su trabajo — no se reactiva sin investigar la causa del loop. / NOC rule: before intervening, read the full STP state. If a port is Blocking, STP did its job — do not reactivate without investigating the loop cause.
Los logs de syslog mostraron la secuencia exacta: Fa0/16 down → Fa0/6 down → reconvergencia. Permiten reconstruir el timeline del incidente. / Syslog showed the exact sequence: Fa0/16 down → Fa0/6 down → reconvergence. Allows reconstructing the incident timeline.	show logging es evidencia forense del incidente. En NOC siempre se captura antes de cerrar el ticket — es la prueba de que el problema ocurrio y fue resuelto. / show logging is forensic evidence. In NOC it is always captured before closing the ticket — it proves the problem occurred and was resolved.

RESULTADO / RESULT

RESULTADO (ES) Loop L2 detectado. STP bloqueo Fa0/16 automaticamente. Cable removido. STP reconvergió. Syslog analizado y documentado como evidencia forense. Ticket NOC-LAB-04 cerrado.	RESULT (EN) L2 loop detected. STP automatically blocked Fa0/16. Cable removed. STP reconverged. Syslog analyzed and documented as forensic evidence. Ticket NOC-LAB-04 closed.
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KEY TAKEAWAY

"STP no es un problema — es la solución. Cuando un puerto está en Blocking, el switch está protegiendo la red de un broadcast storm. En NOC, antes de 'arreglar' un puerto bloqueado hay que entender POR QUÉ está bloqueado. El syslog es la evidencia forense que permite reconstruir el timeline del incidente y cerrar el ticket con pruebas concretas."

"STP is not the problem — it is the solution. When a port is in Blocking state, the switch is protecting the network from a broadcast storm. In NOC, before 'fixing' a blocked port you must understand WHY it is blocked. The syslog is the forensic evidence that allows reconstructing the incident timeline and closing the ticket with concrete proof."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining environment (e.g., Newmont Cerro Negro), L2 loops are one of the most dangerous network events — a broadcast storm can saturate the entire LAN segment in seconds, taking down SCADA communications, IP cameras, and safety systems simultaneously. A NOC engineer must: (1) recognize the symptoms (high CPU on switch, broadcast traffic spikes), (2) verify STP is active and blocking the correct port, (3) identify the physical cause of the redundant path (miscabling, unauthorized device), (4) remove the loop source, and (5) document the full incident timeline using show logging before closing the ticket. Never re-enable a blocked port without understanding why STP blocked it.